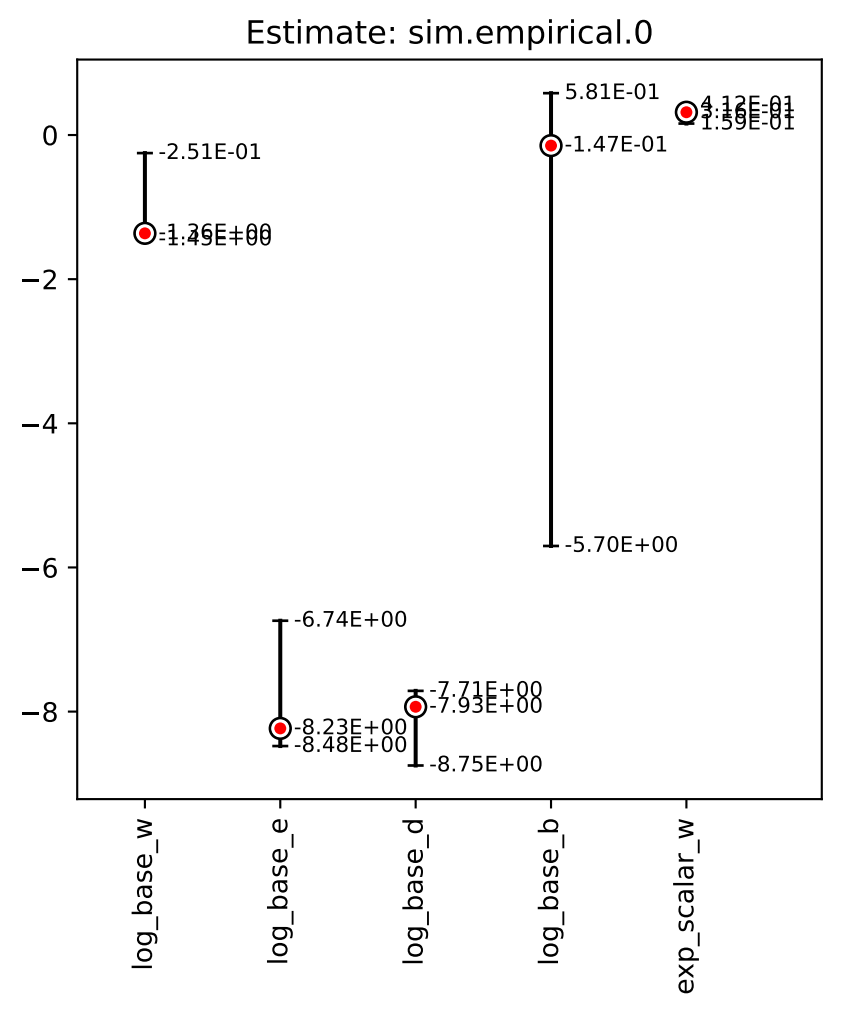
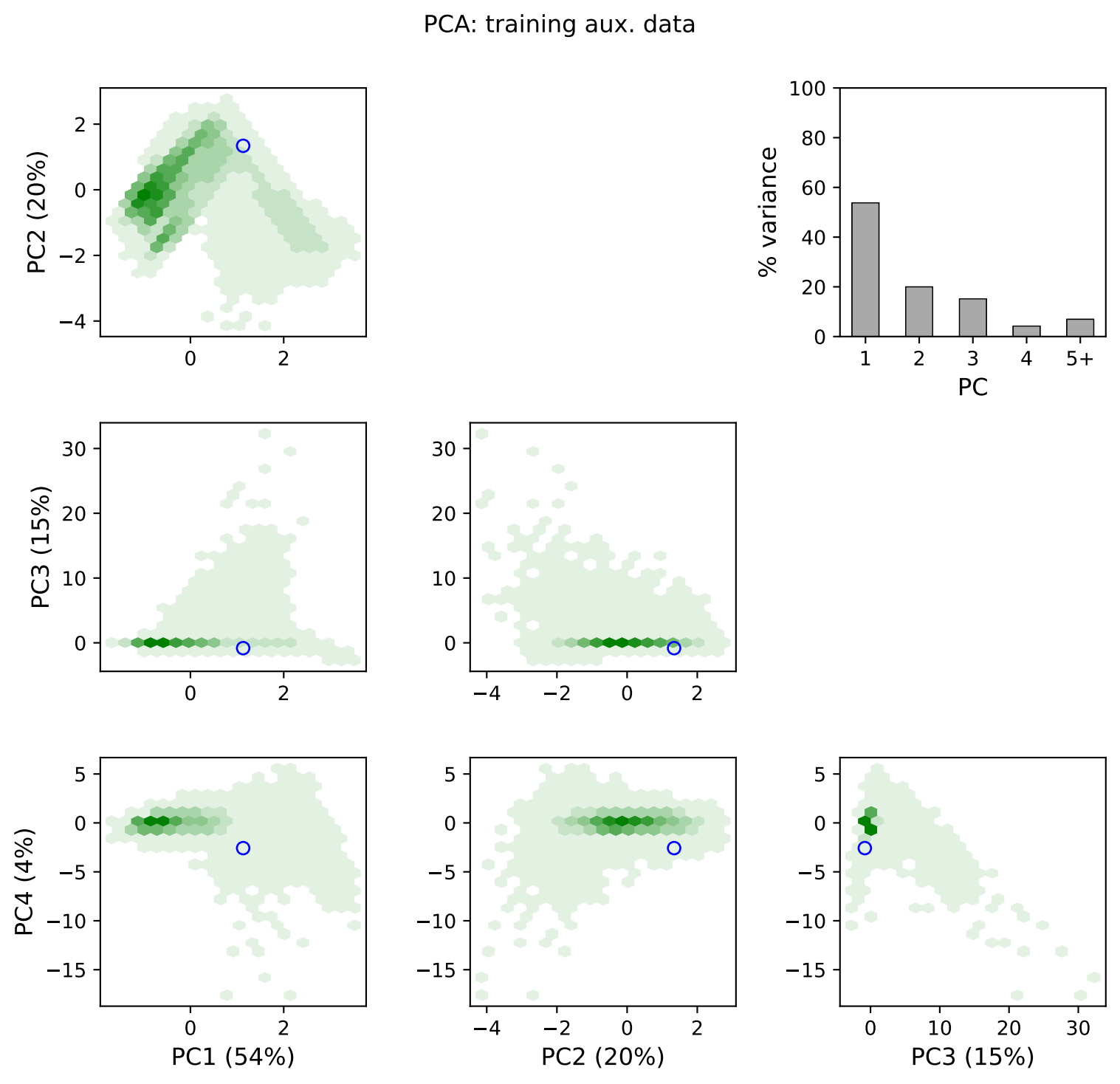
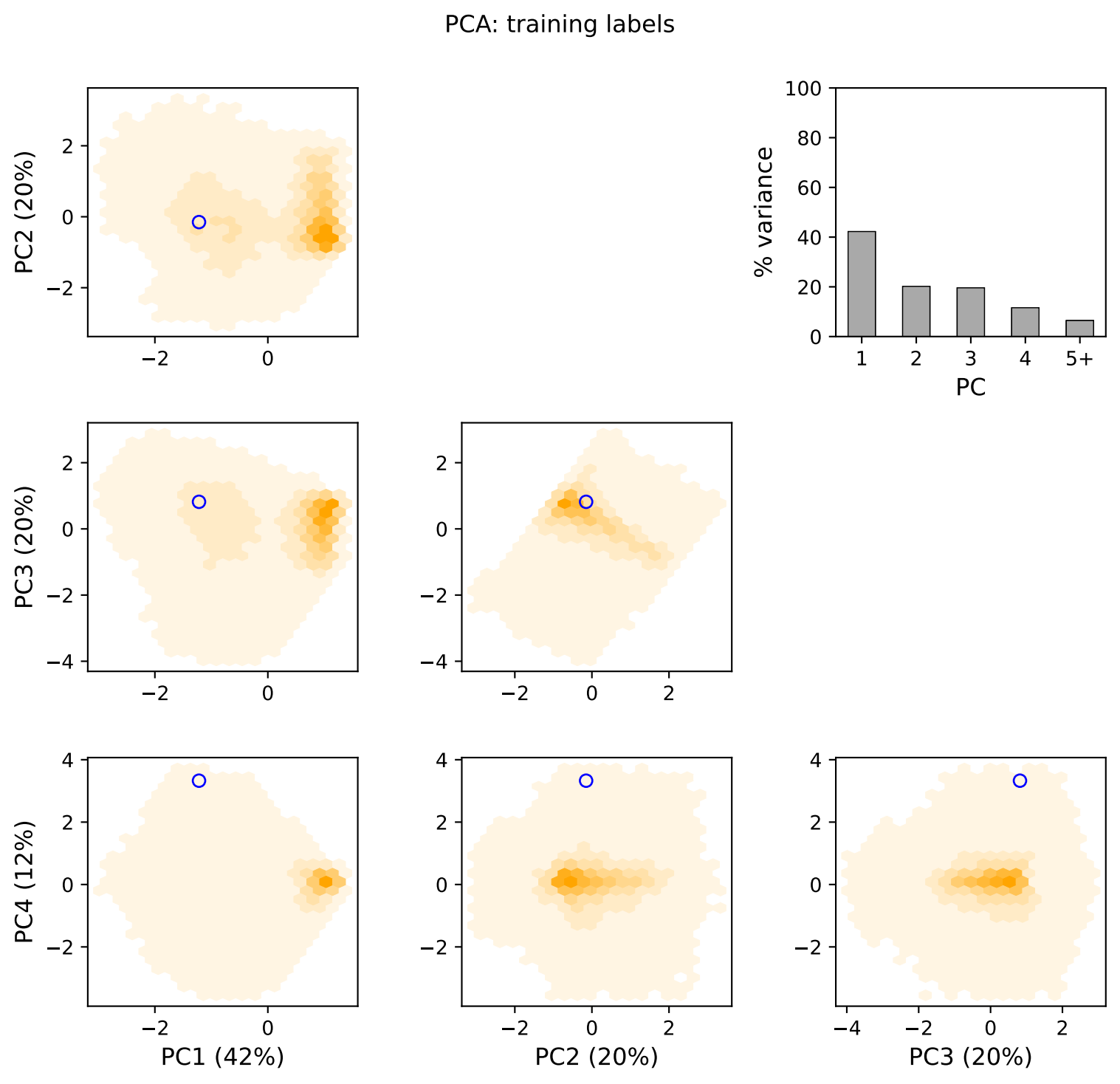
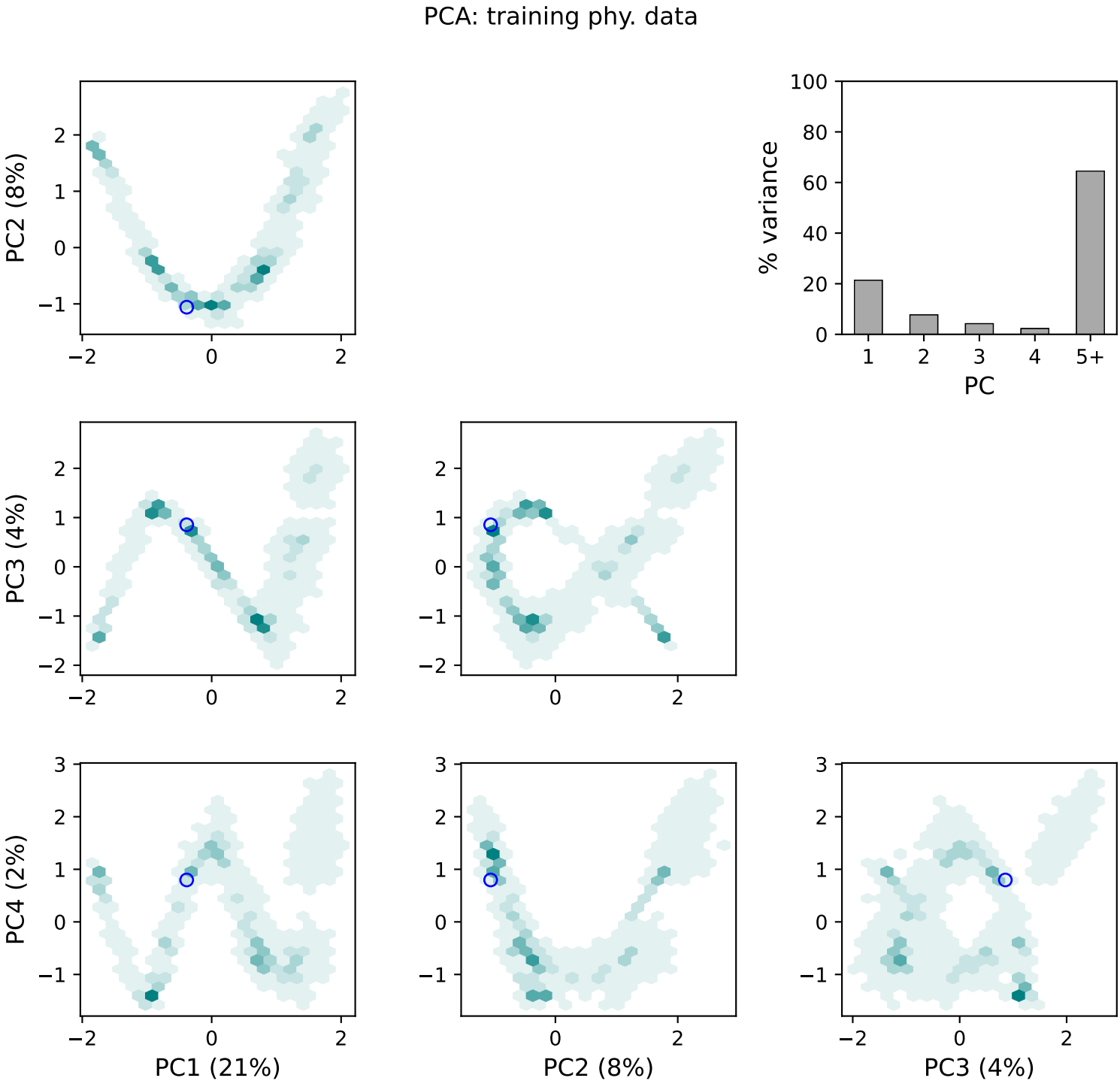
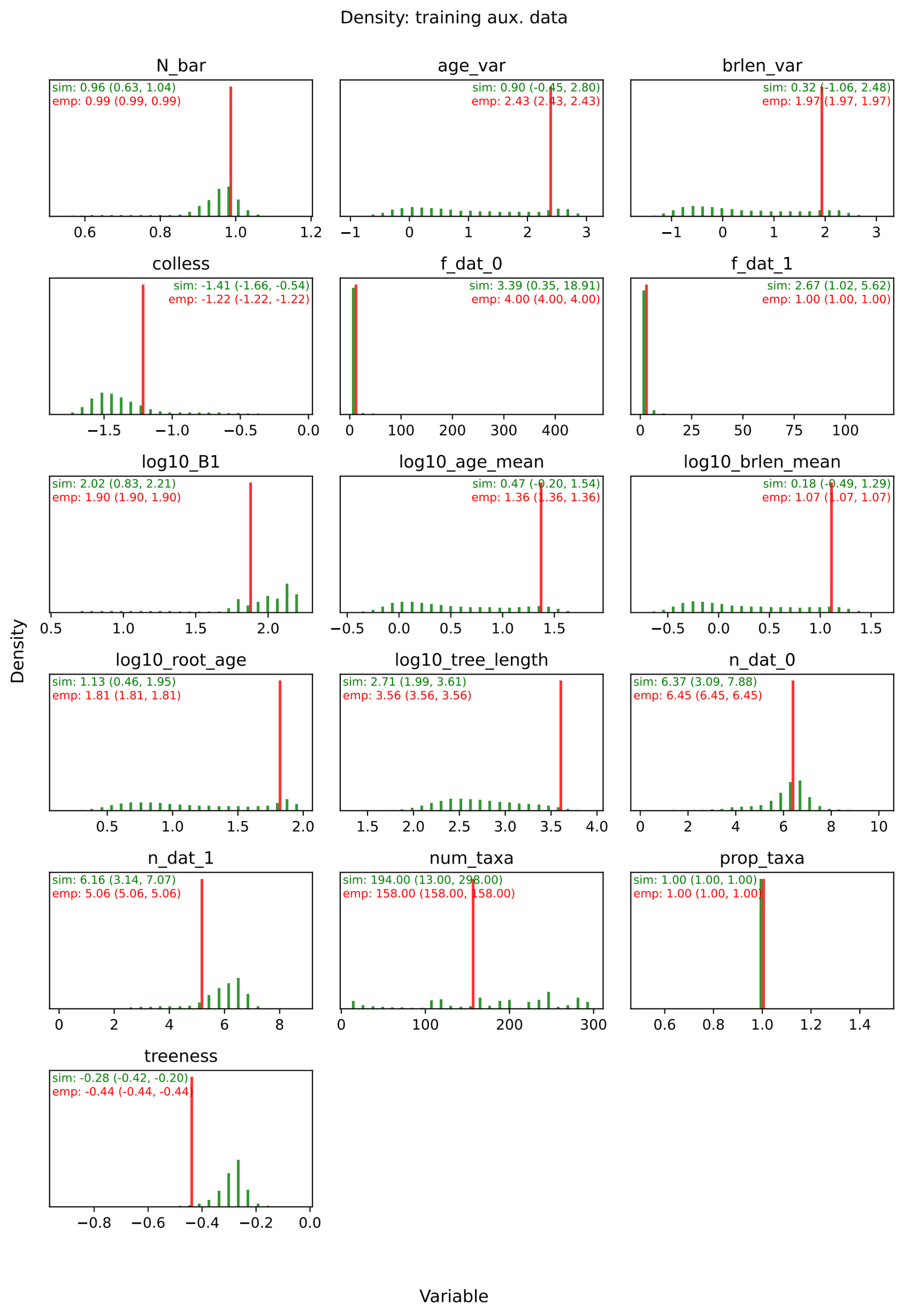
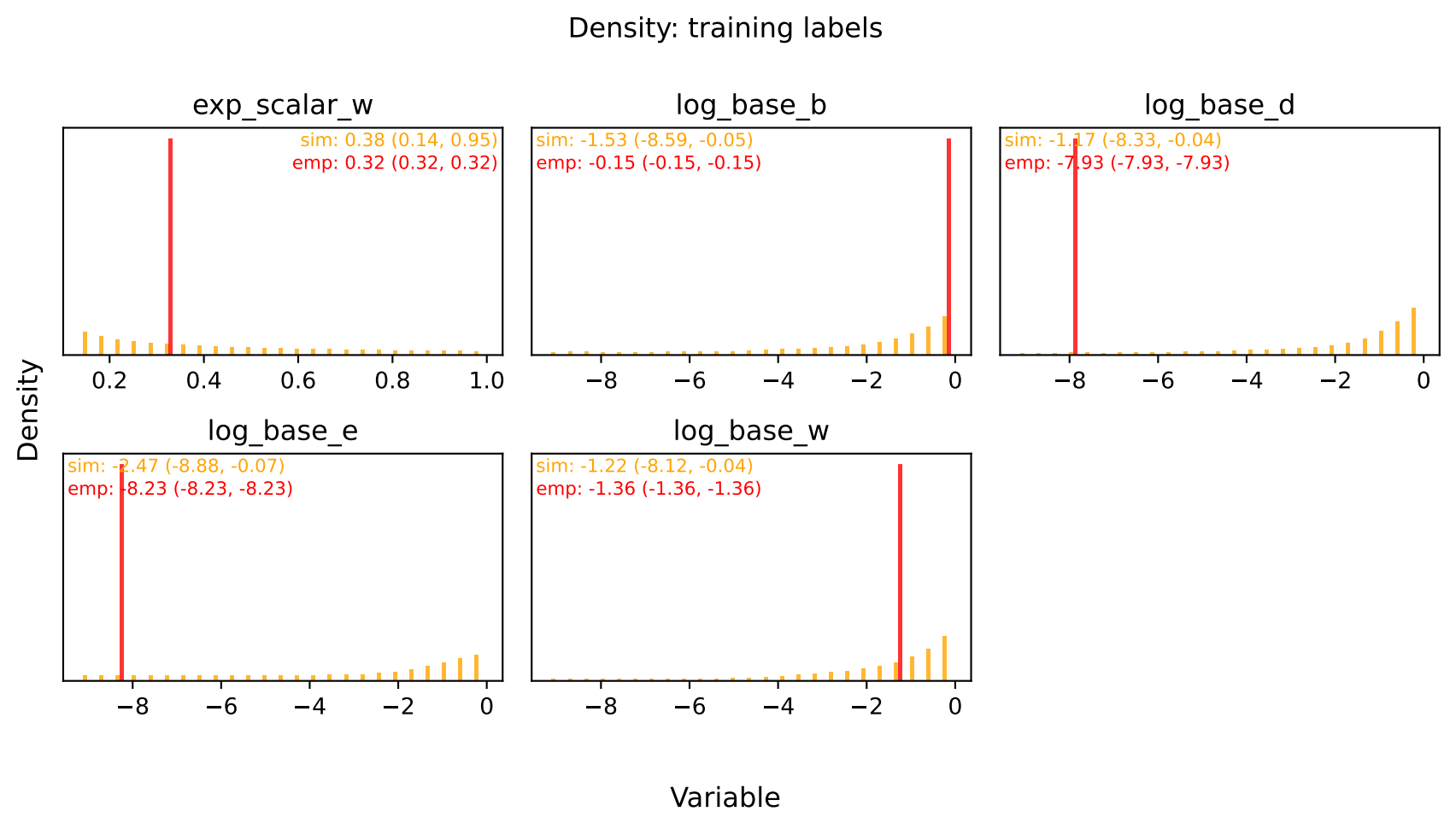

Estimate: sim.empirical.0
0
−2
−4
−6
−8
-2.51E-01
-1.26E+00
-1.45E+00
5.81E-01
-1.47E-01
-5.70E+00
4.12E-01
1.59E-01
-6.74E+00
-8.23E+00
-8.48E+00
-7.71E+00
-7.93E+00
-8.75E+00
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w

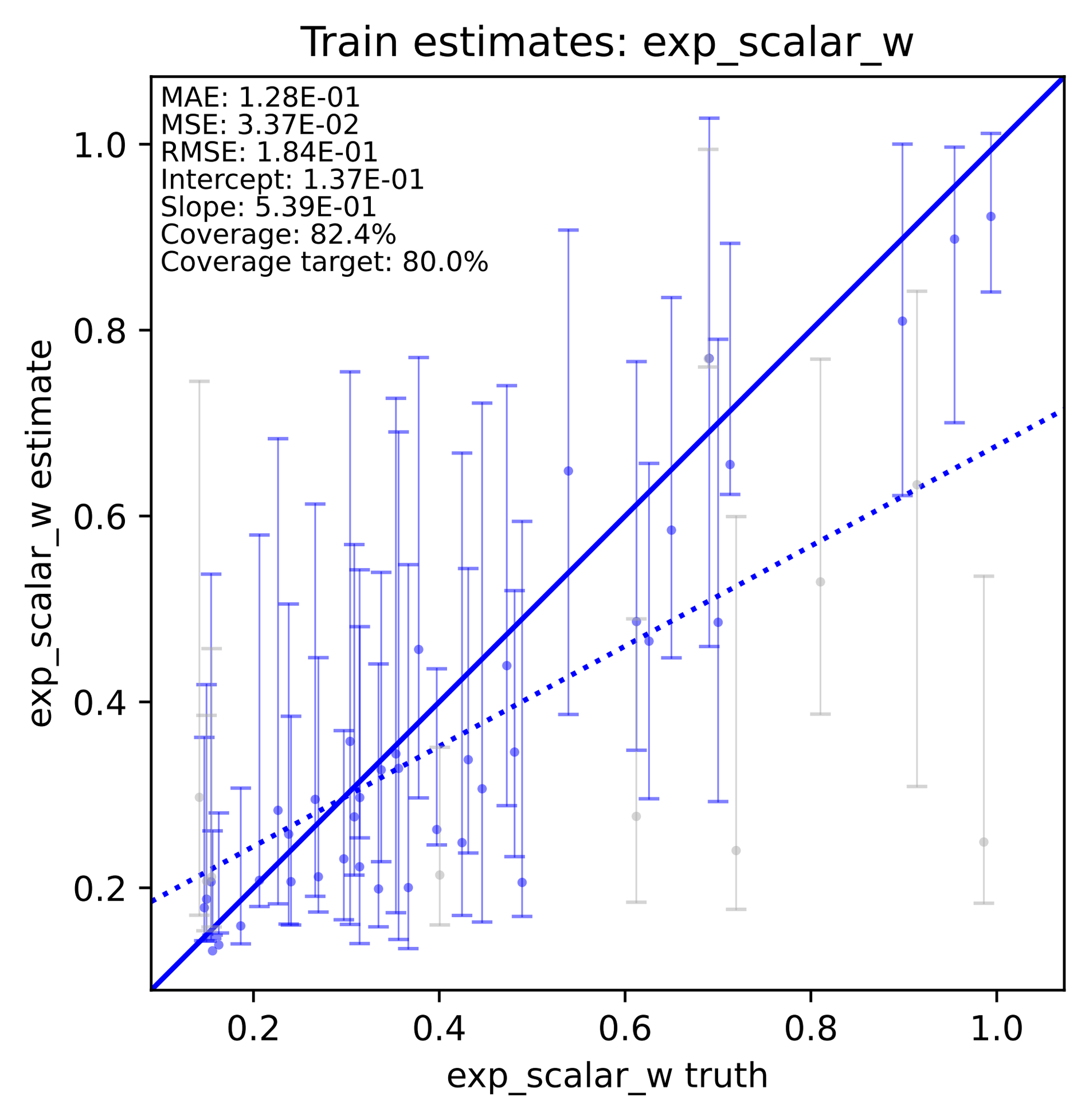

PCA: training aux. data
PC2 (20%)
% variance
PC
PC3 (15%)
PC4 (4%)
PC1 (54%)
PC2 (20%)
PC3 (15%)

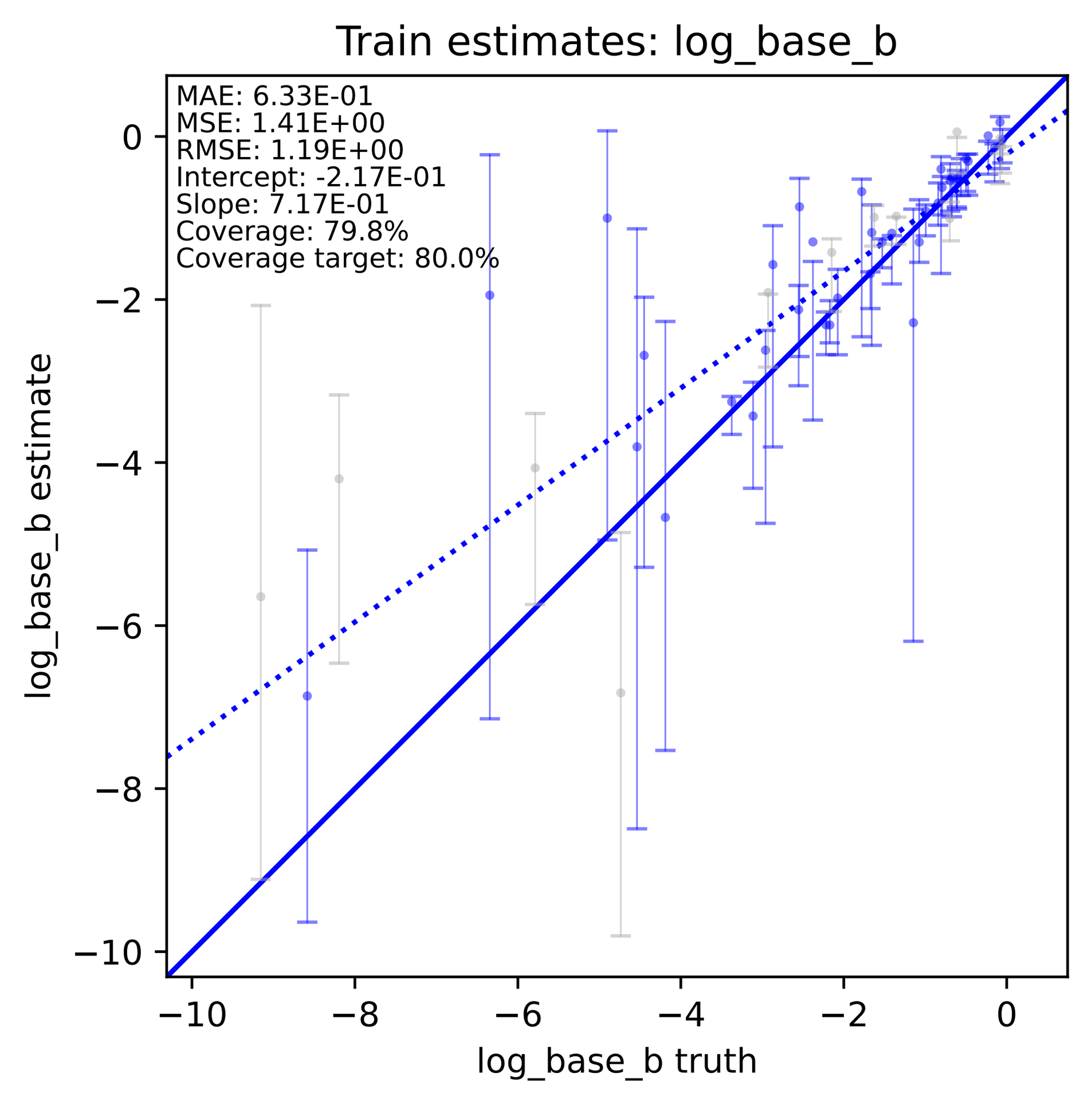

PCA: training labels
PC2 (20%)
% variance
PC
PC3 (20%)
PC4 (12%)
PC1 (42%)
PC2 (20%)
PC3 (20%)

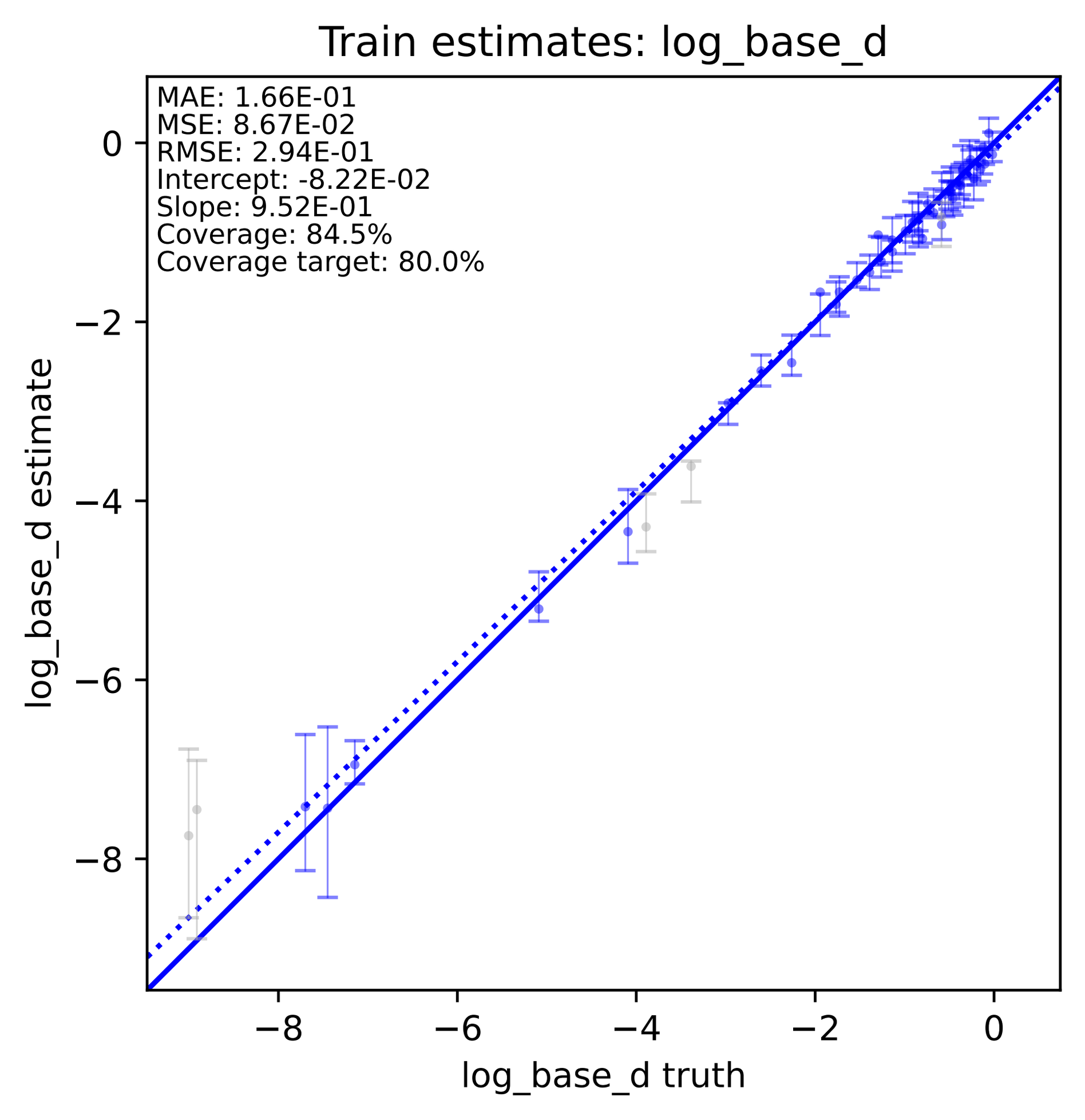

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (21%)
PC2 (8%)
PC3 (4%)

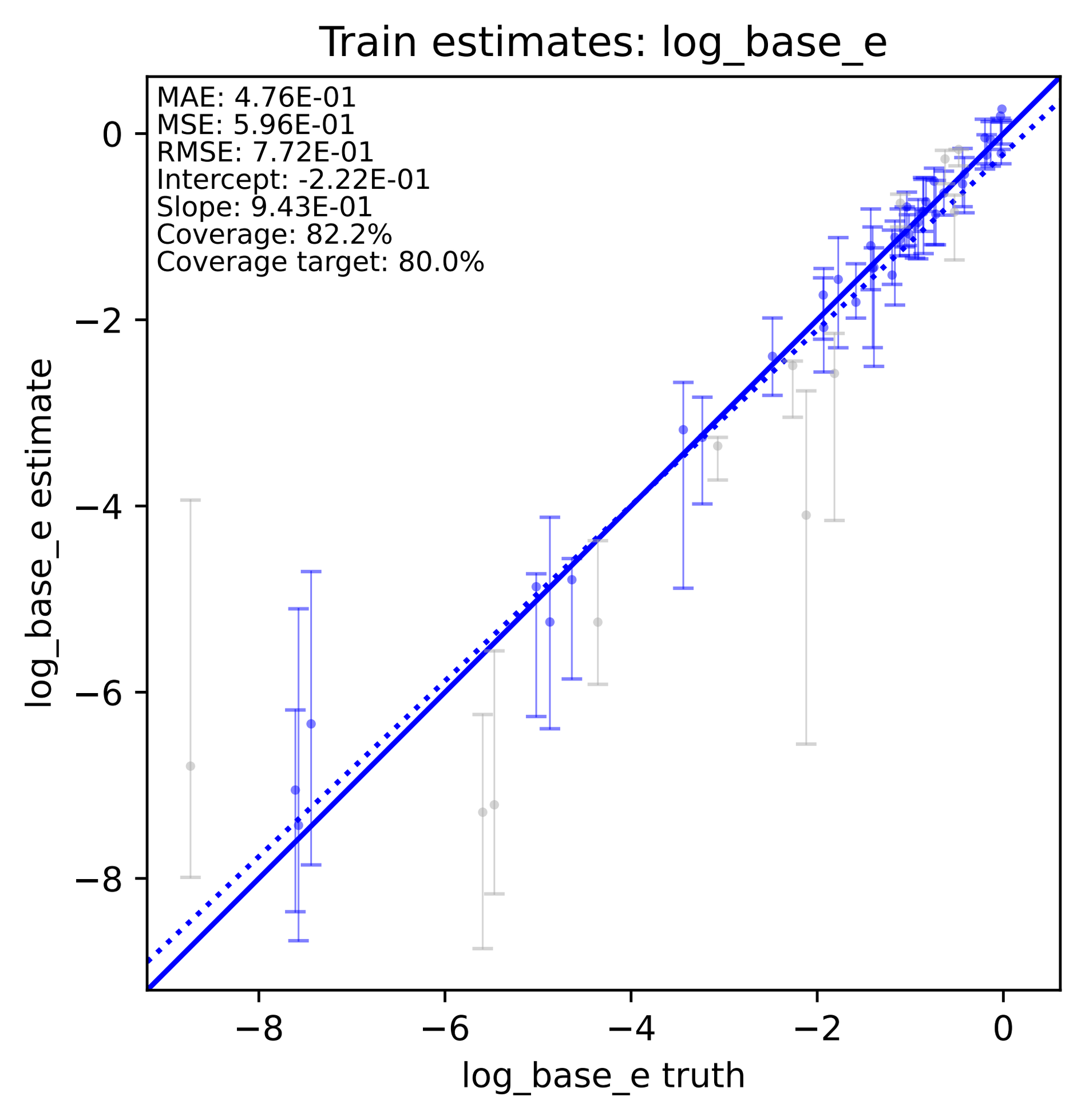

Density: training aux. data
N_bar
sim: 0.96 (0.63, 1.04)
emp: 0.99 (0.99, 0.99)
age_var
sim: 0.90 (-0.45, 2.80)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.32 (-1.06, 2.48)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.41 (-1.66, -0.54)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 3.39 (0.35, 18.91)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 2.67 (1.02, 5.62)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 2.02 (0.83, 2.21)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.47 (-0.20, 1.54)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.18 (-0.49, 1.29)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.13 (0.46, 1.95)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.71 (1.99, 3.61)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.37 (3.09, 7.88)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 6.16 (3.14, 7.07)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 194.00 (13.00, 298.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.42, -0.20)
emp: -0.44 (-0.44, -0.44)
Density
Variable

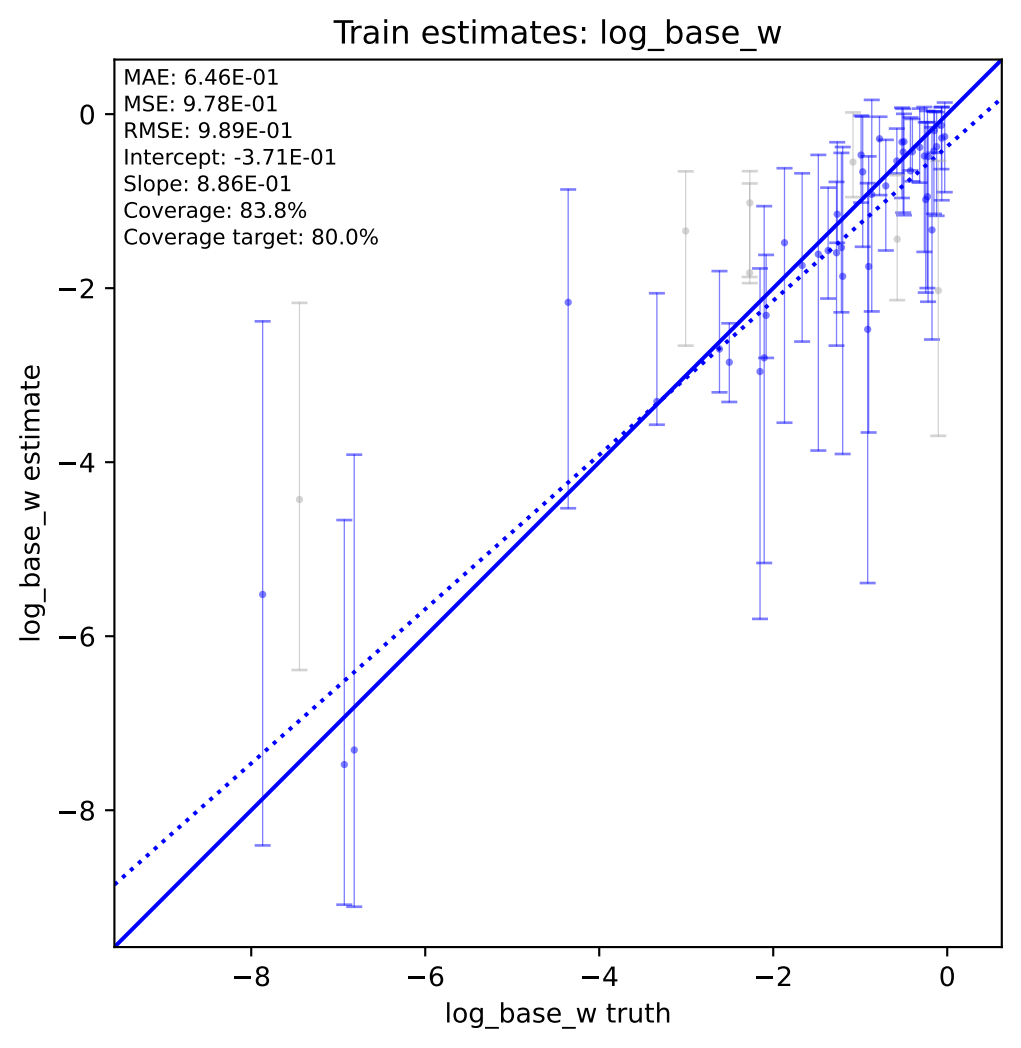

Density: training labels
exp_scalar_w
sim: 0.38 (0.14, 0.95)
emp: 0.32 (0.32, 0.32)
log_base_b
sim: -1.53 (-8.59, -0.05)
emp: -0.15 (-0.15, -0.15)
log_base_d
sim: -1.17 (-8.33, -0.04)
emp: -7.93 (-7.93, -7.93)
log_base_e
sim: -2.47 (-8.88, -0.07)
emp: -8.23 (-8.23, -8.23)
log_base_w
sim: -1.22 (-8.12, -0.04)
emp: -1.36 (-1.36, -1.36)
Density
Variable

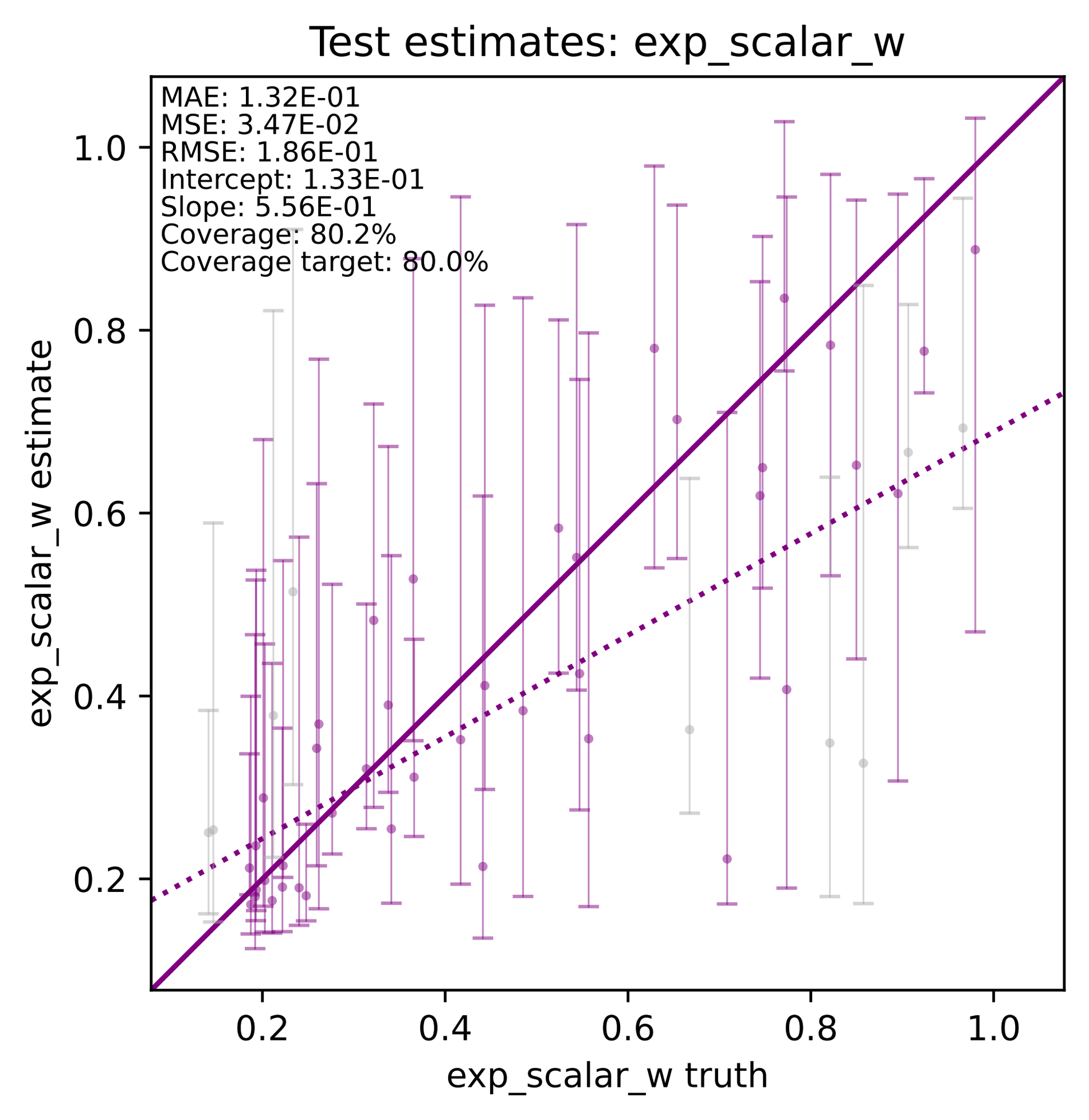

Train estimates: exp_scalar_w
MAE: 1.28E-01
MSE: 3.37E-02
RMSE: 1.84E-01
Intercept: 1.37E-01
Slope: 5.39E-01
Coverage: 82.4%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

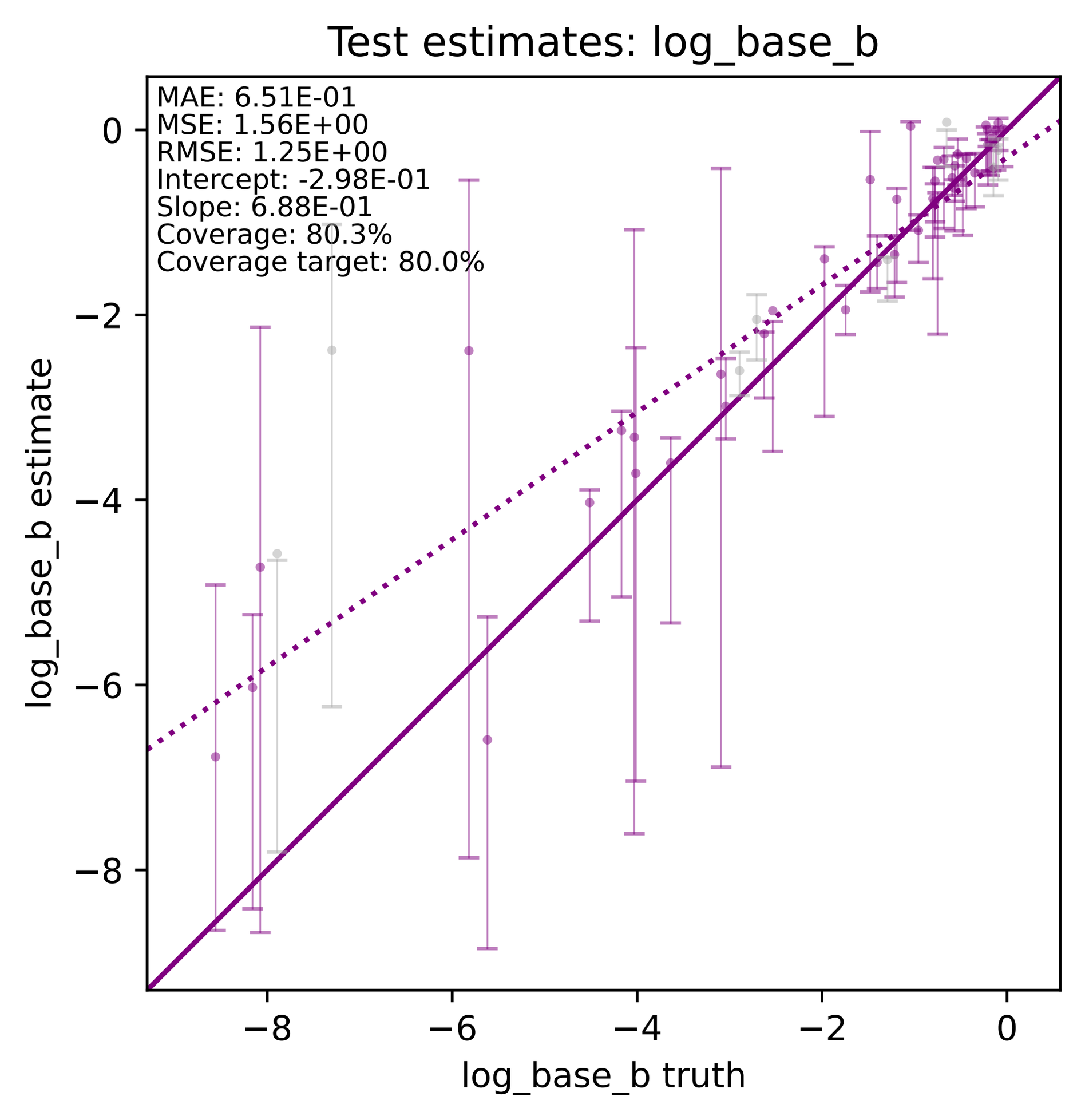

Train estimates: log_base_b
MAE: 6.33E-01
MSE: 1.41E+00
RMSE: 1.19E+00
Intercept: -2.17E-01
Slope: 7.17E-01
Coverage: 79.8%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

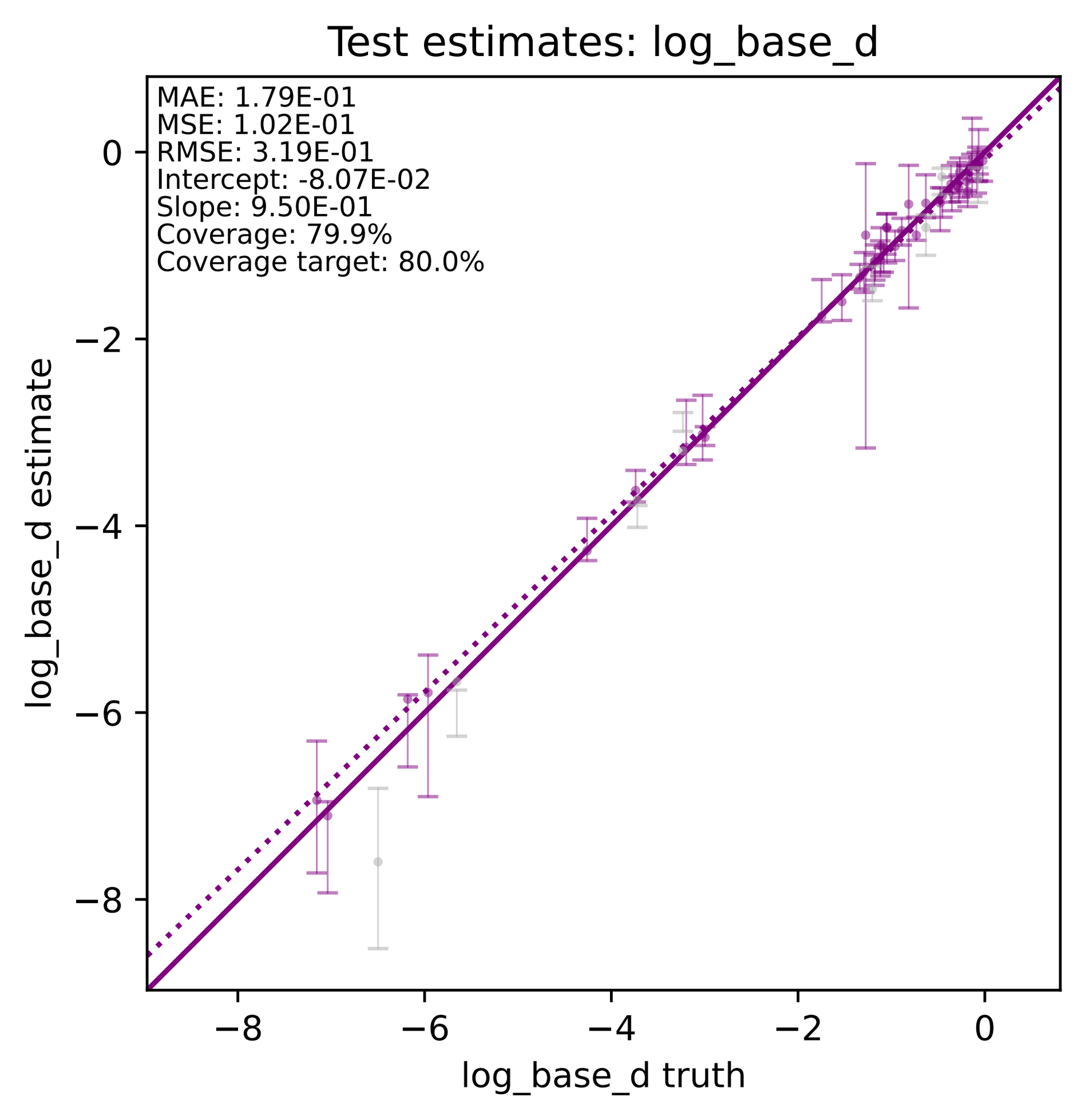

Train estimates: log_base_d
MAE: 1.66E-01
MSE: 8.67E-02
RMSE: 2.94E-01
Intercept: -8.22E-02
Slope: 9.52E-01
Coverage: 84.5%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

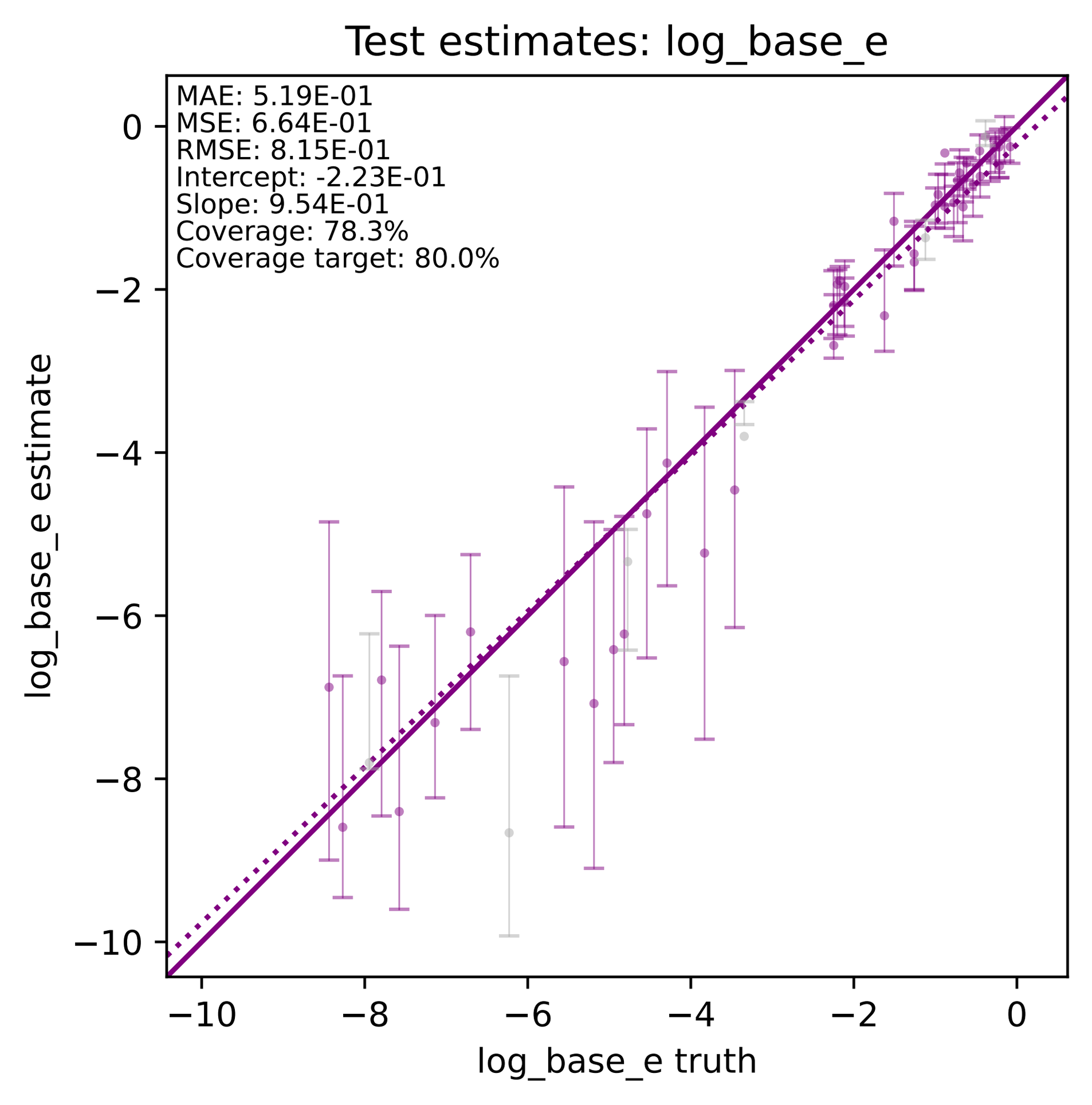

Train estimates: log_base_e
MAE: 4.76E-01
MSE: 5.96E-01
RMSE: 7.72E-01
Intercept: -2.22E-01
Slope: 9.43E-01
Coverage: 82.2%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

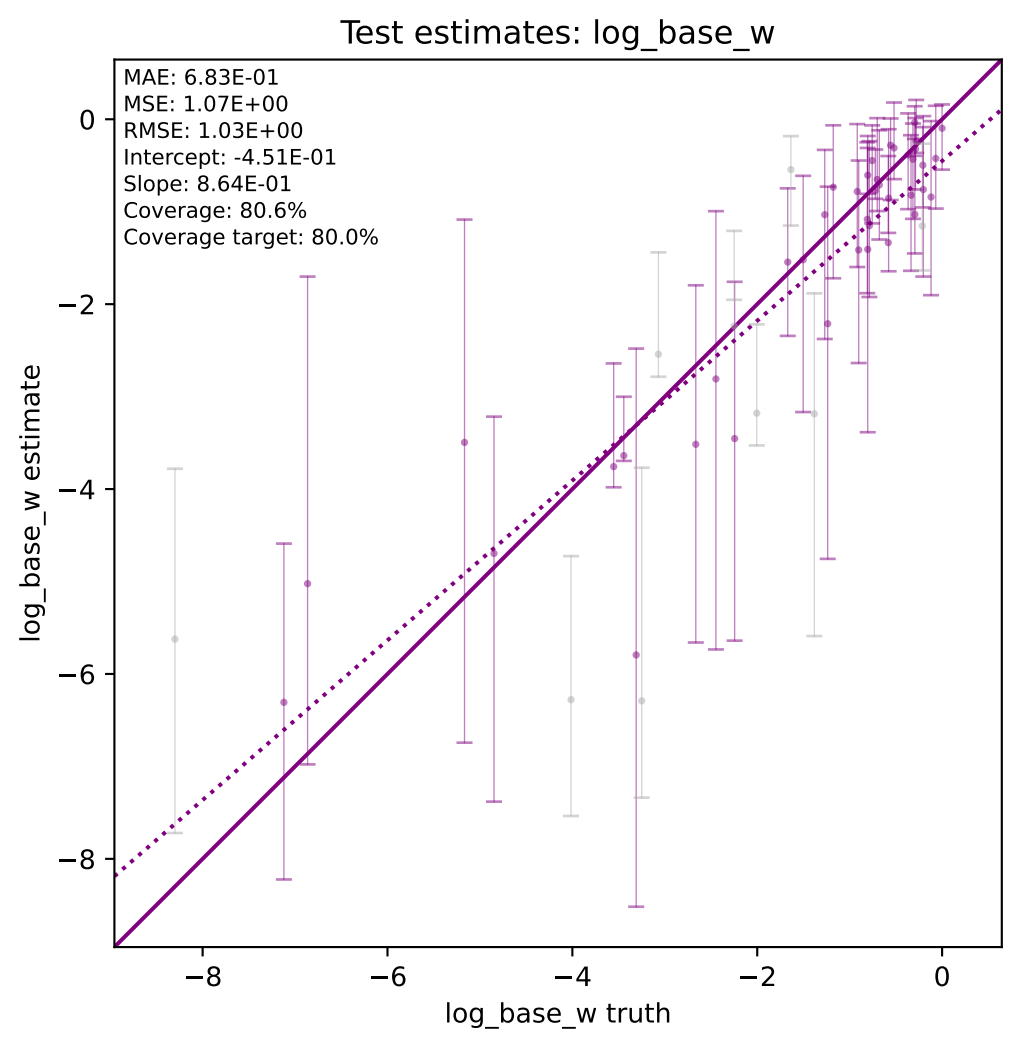

Train estimates: log_base_w
MAE: 6.46E-01
MSE: 9.78E-01
RMSE: 9.89E-01
Intercept: -3.71E-01
Slope: 8.86E-01
Coverage: 83.8%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

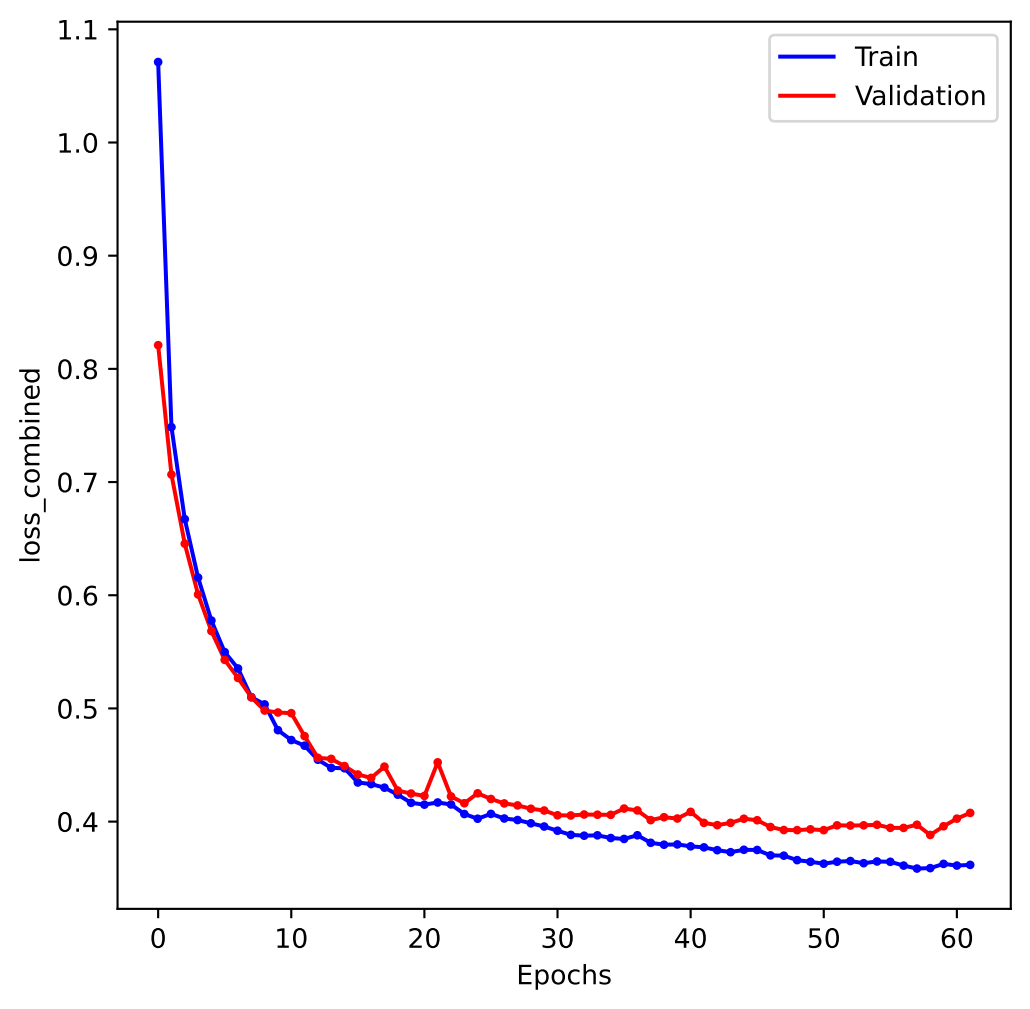

Test estimates: exp_scalar_w
MAE: 1.32E-01
MSE: 3.47E-02
RMSE: 1.86E-01
Intercept: 1.33E-01
Slope: 5.56E-01
Coverage: 80.2%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

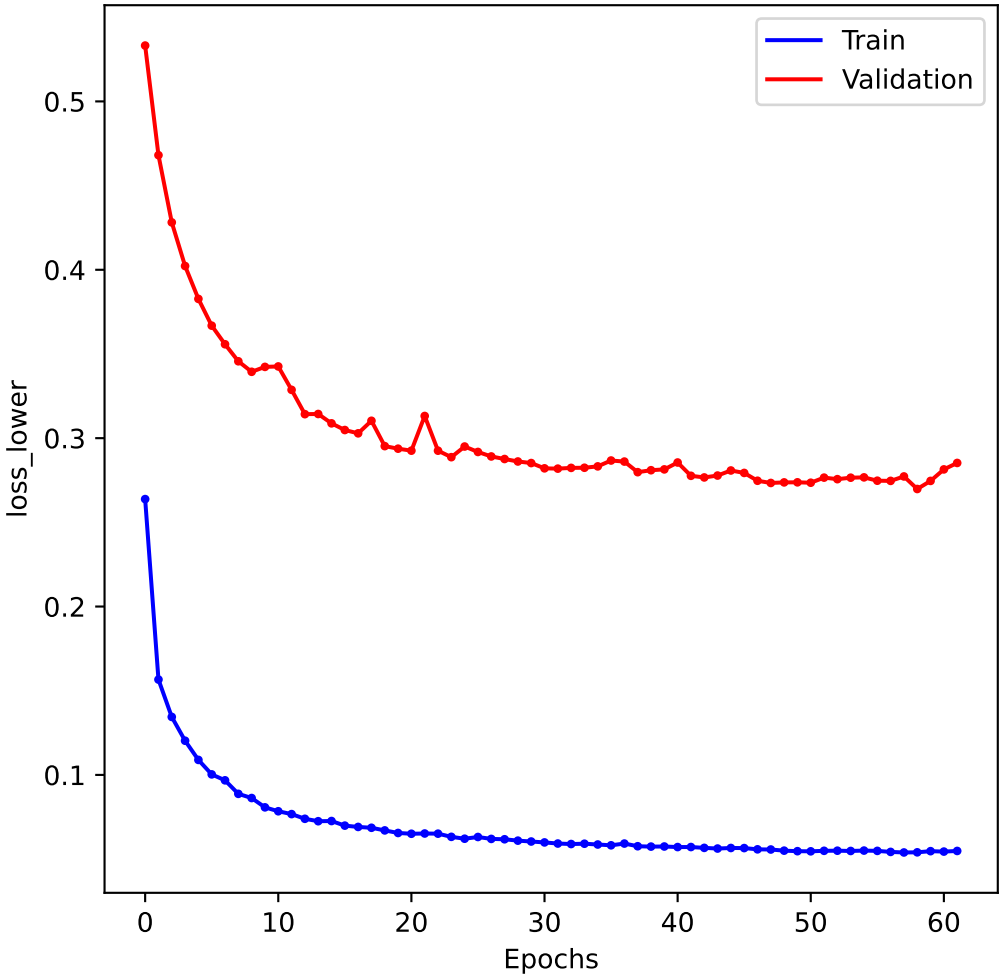

Test estimates: log_base_b
MAE: 6.51E-01
MSE: 1.56E+00
RMSE: 1.25E+00
Intercept: -2.98E-01
Slope: 6.88E-01
Coverage: 80.3%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

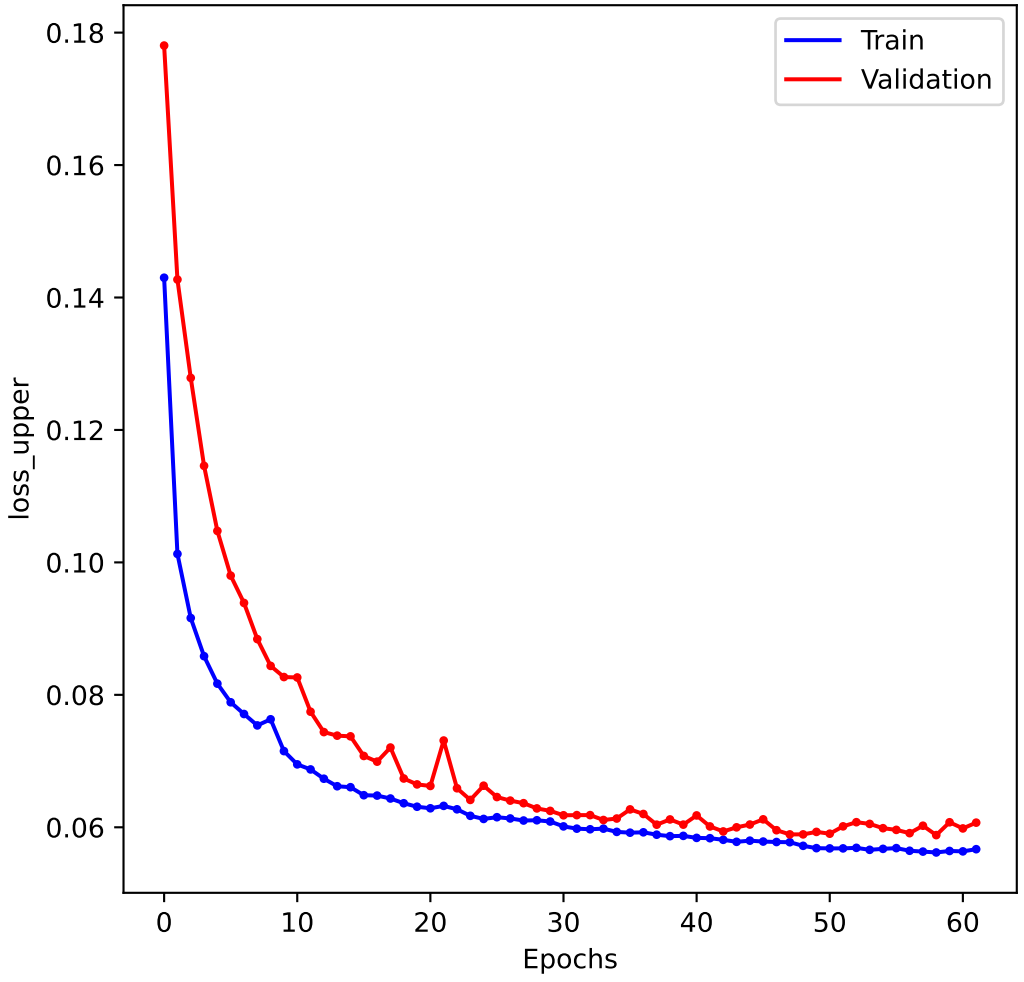

Test estimates: log_base_d
MAE: 1.79E-01
MSE: 1.02E-01
RMSE: 3.19E-01
Intercept: -8.07E-02
Slope: 9.50E-01
Coverage: 79.9%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

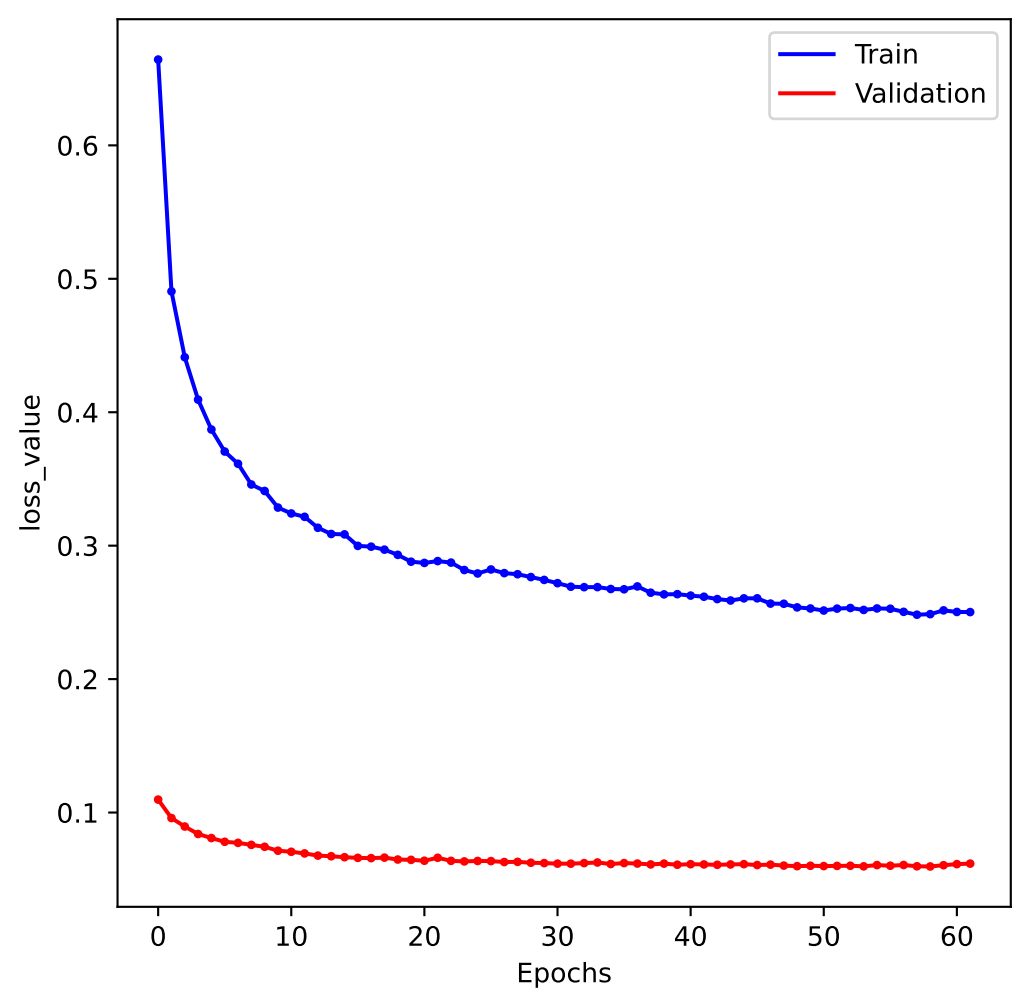

Test estimates: log_base_e
MAE: 5.19E-01
MSE: 6.64E-01
RMSE: 8.15E-01
Intercept: -2.23E-01
Slope: 9.54E-01
Coverage: 78.3%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

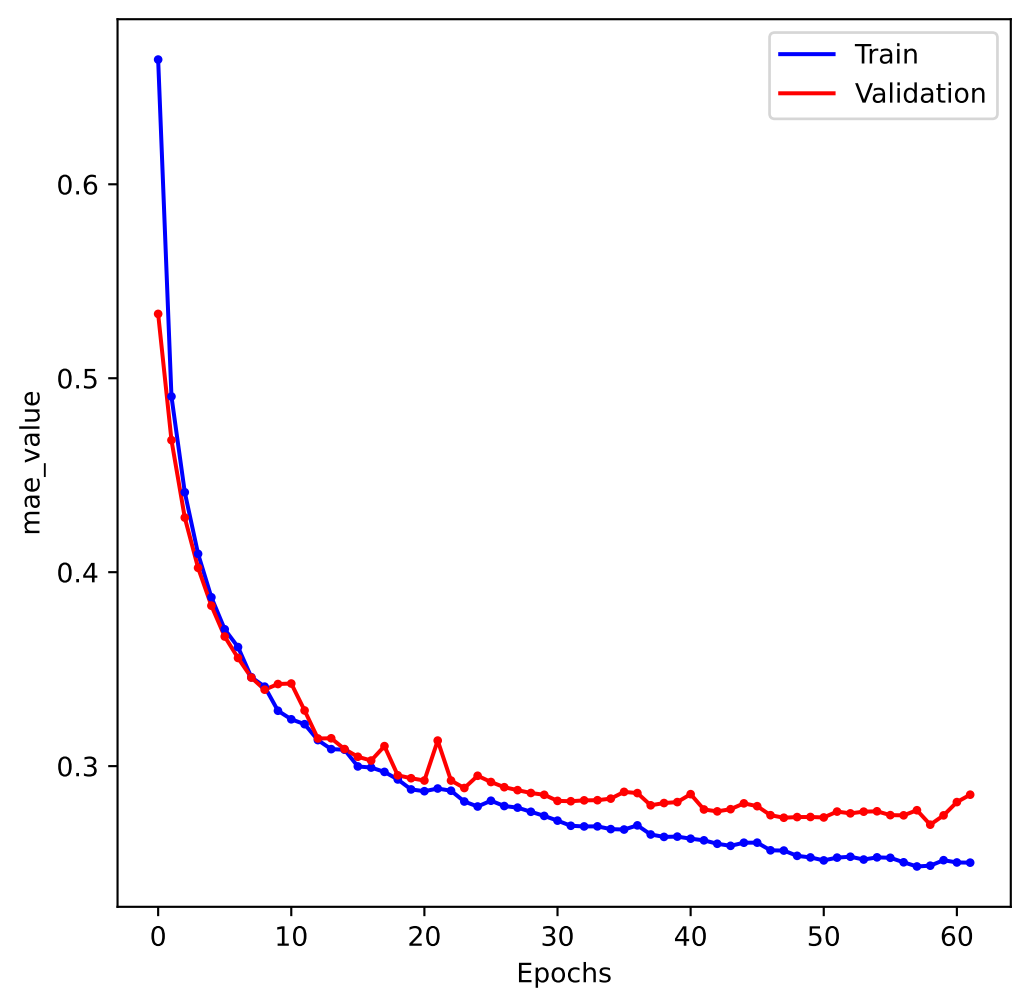

Test estimates: log_base_w
MAE: 6.83E-01
MSE: 1.07E+00
RMSE: 1.03E+00
Intercept: -4.51E-01
Slope: 8.64E-01
Coverage: 80.6%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

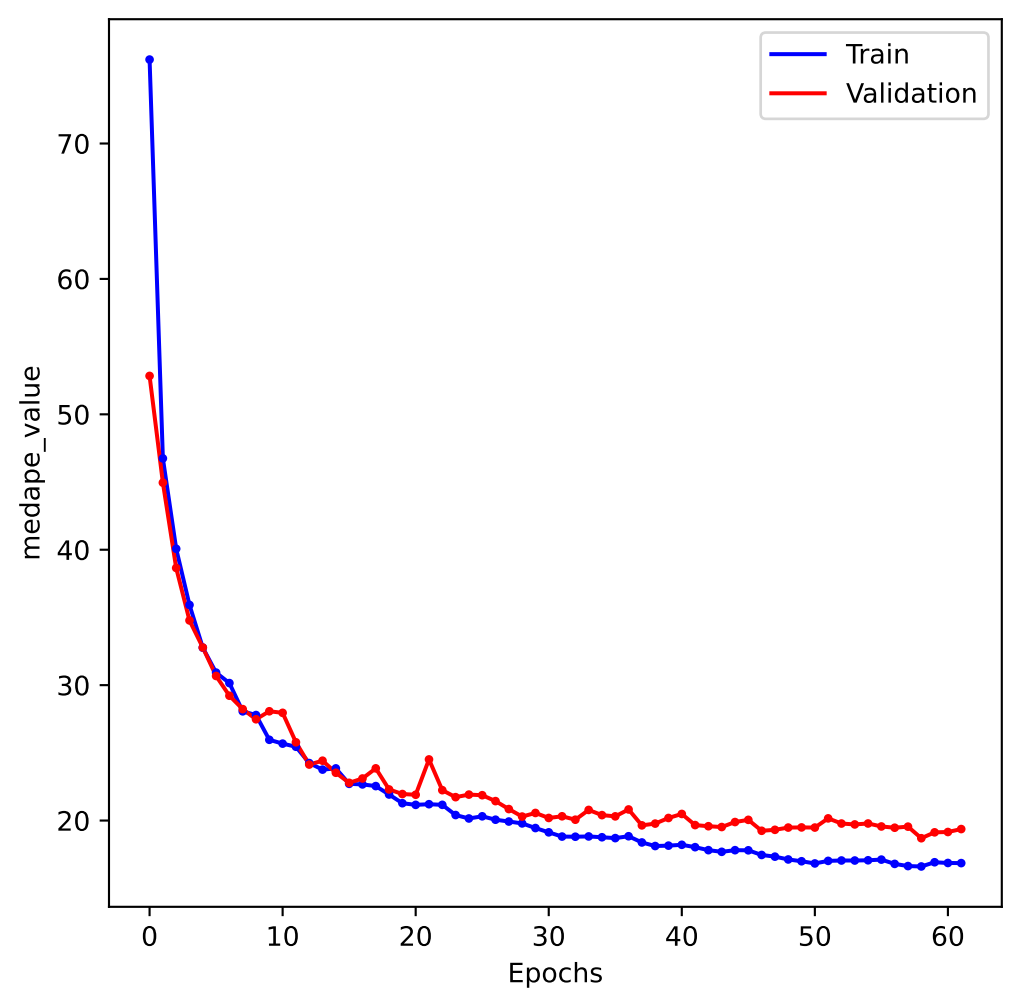

Train
Validation
loss_combined
Epochs
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50
60

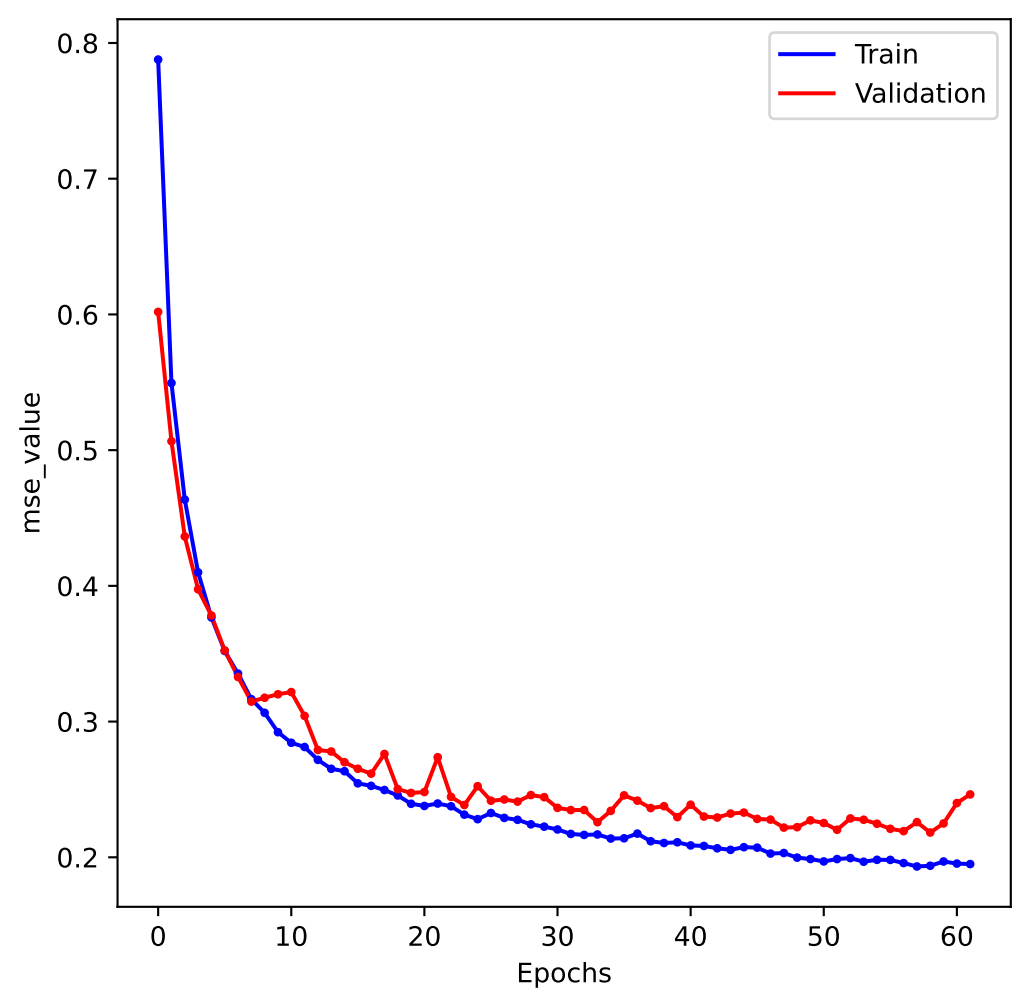

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
Epochs

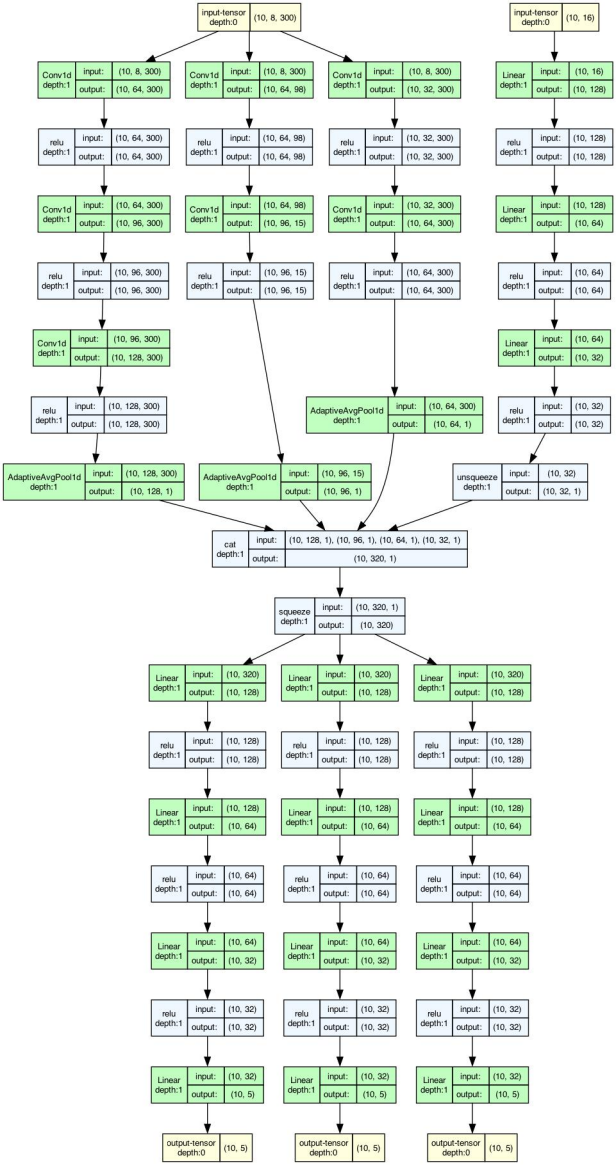

Train
Validation
loss_upper
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0
10
20
30
40
50
60
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
Epochs


Train
Validation
mae_value
0.6
0.5
0.4
0.3
0
10
20
30
40
50
60
Epochs


Train
Validation
medape_value
70
60
50
40
30
20
0
10
20
30
40
50
60
Epochs


Train
Validation
mse_value
Epochs
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
10
20
30
40
50
60


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)